

BITS PILANI, DUBAI CAMPUS

Dubai International Academic City

Academic-Graduate Studies and Research Division

Second Semester 2020 – 2021

Comprehensive Examination (Closed Book)

Year : M.E. (HD)

Date: 27 May 2020

Time: 7:00pm – 10.00pm

Course No : CS F464

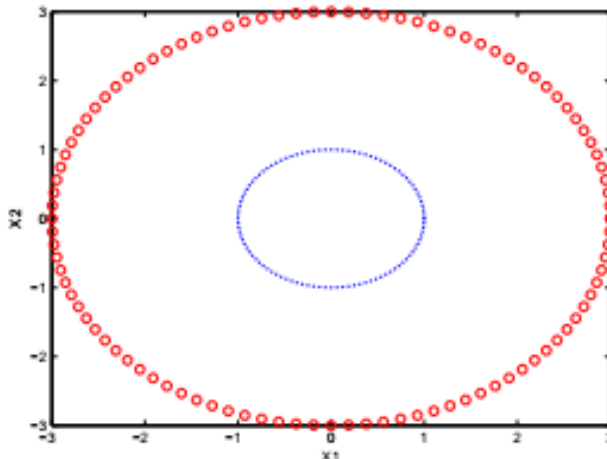
MAX Marks: Part A:20 M, Part B 20 M Weightage: (40%)

Course Title : Machine Learning

Part: A

Duration: 3 Hours

Part A: Answer all questions. Show the steps required to arrive at the answer for questions.

Q. A1	Calculate the best LDA (Linear Discriminant Analysis) projection vector for following dataset. (Show all steps of calculations) $X_1 = (x_1, x_2) = \{(1,4), (3,7), (2,4)\}$ $X_2 = (x_1, x_2) = \{(11,10), (9,11), (8,9)\}$	6 M										
Q. A2	Consider a following Figure 1 which represents the dataset which consist of two classes (Assumptions: Classes have equal number of instances). Draw the decision boundary if we do training using SVM with three different kernels, linear, polynomial and RBF.  Figure 1 Q. A2	3 M										
Q. A3	Differentiate between k-means clustering and Gaussian mixture with expectation maximization clustering. (Minimum 3 correct differences)	3 M										
Q. A4	Given a following observed datapoints ($X: x_i$) with two classes A and B, <table border="1" data-bbox="475 1632 1042 1823"><thead><tr><th>Class A</th><th>Class B</th></tr></thead><tbody><tr><td>10</td><td>14</td></tr><tr><td>11</td><td>11</td></tr><tr><td>8</td><td>9</td></tr><tr><td>10</td><td>13</td></tr></tbody></table> The prior probabilities of class A and B are 0.4 and 0.6 respectively. What will be the class of following unlabeled data points (x_1 and x_2) using Gaussian for classification? (Show all steps of calculations) i. $x_1 = 5$ ii. $x_2 = 12$	Class A	Class B	10	14	11	11	8	9	10	13	4 M
Class A	Class B											
10	14											
11	11											
8	9											
10	13											

Q. A5

Consider a neural network with linear activation as shown in Figure 2. The output of each unit is a constant C multiplied by the weighted sum of inputs.

4 M

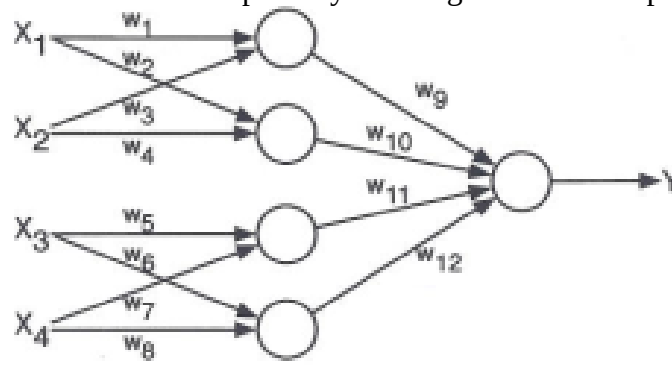


Figure 2 Q. A5

- i. Represent the above network using single unit network, including the weights and activation function.
- ii. Can it be possible to represent the above network using linear regression? If yes, write the equation of Y . If no, explain why?

***** ALL THE BEST *****

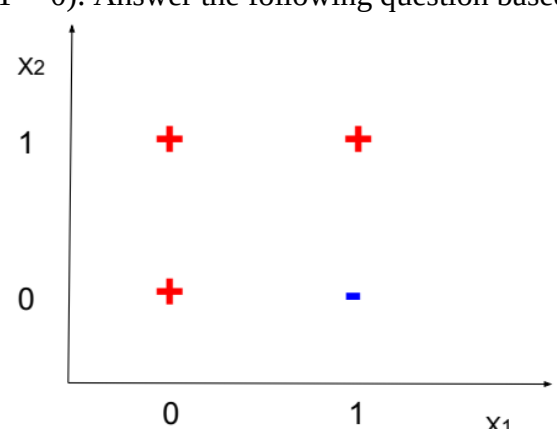
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Comprehensive Examination (Closed Book)**Year** : M.E. (HD)**Date:** 27 May 2020**Time:** 7:00pm – 10.00pm**Course No** : CS F464**MAX Marks:** Part A:20 M, Part B 20 M Weightage: (40%)**Course Title** : Machine Learning**Part:** B**Duration:** 3 Hours**Part B:** Answer all questions. Show the steps required to arrive at the answer for questions.

Q. B1	Consider a census data represented in Figure 1. From given data, we want to predict either person earn more than 20K a year. For clarity we modelled the two features with boolean variables $X_1, X_2 \in \{0, 1\}$ and $Y = 1$ indicate that a person earns more than 20K a year. The Figure 1 shows three positive samples ($Y = 1$) and one negative samples ($Y = 0$). Answer the following question based on given data.  Figure 1 Q. B1 - Is there any logistic regression classifier using X_1 and X_2 that can perfectly classify the examples in Figure 1? How about if we change label of point (0,1) from “+” to “-”? - If we train the Navie Bayes classifier using maximum likelihood estimation. Calculate the prior probability $P(Y)$ and conditional probabilities $P(X_i Y = 1)$.	5 M																					
Q. B2	Differentiate between active learning, supervised learning and semi-supervised learning. (Minimum 3 correct differences)	3 M																					
Q. B3	Consider a following dataset which shows willingness of certain customer to eat a chocolate of certain manufactures with specific flavor, <table border="1" data-bbox="379 1680 1141 1948"> <thead> <tr> <th>Manufacturer</th><th>Flavor</th><th>Willingness to eat</th></tr> </thead> <tbody> <tr> <td>Lindt</td><td>Strawberry</td><td>Yes</td></tr> <tr> <td>Lindt</td><td>Milk</td><td>Yes</td></tr> <tr> <td>Cadbury</td><td>Strawberry</td><td>Yes</td></tr> <tr> <td>Cadbury</td><td>Milk</td><td>No</td></tr> <tr> <td>Milka</td><td>Strawberry</td><td>No</td></tr> <tr> <td>Milka</td><td>Milk</td><td>No</td></tr> </tbody> </table> In above dataset, which feature (Manufacturer or flavor) has the larger information gain with willingness to eat?	Manufacturer	Flavor	Willingness to eat	Lindt	Strawberry	Yes	Lindt	Milk	Yes	Cadbury	Strawberry	Yes	Cadbury	Milk	No	Milka	Strawberry	No	Milka	Milk	No	4 M
Manufacturer	Flavor	Willingness to eat																					
Lindt	Strawberry	Yes																					
Lindt	Milk	Yes																					
Cadbury	Strawberry	Yes																					
Cadbury	Milk	No																					
Milka	Strawberry	No																					
Milka	Milk	No																					

CS F464 Machine Learning Question Paper: Part-B 27.05.2021, 2 pages, 5 questions

	ii. Also draw the decision tree for predicting willingness to eat which maximizes the information gain?										
Q. B4	Calculate the Q-learning table for following listed state, rewards, and actions sequences for two states and two action system. i. Current state: S_2, Reward (R) = +20, Action (a_1): $S_2 \rightarrow S_1$ ii. Current state: S_1, Reward (R) = -10, Action (a_2): $S_1 \rightarrow S_2$ The learning rate (α) is 0.5 and a discount factor (γ) is 0.5 for each step. The initial Q-table is as follows, <table><tr><th><u>Q-values</u></th><th>S1</th><th>S2</th></tr><tr><th>a1</th><td>-5</td><td>0</td></tr><tr><th>a2</th><td>-5</td><td>0</td></tr></table>	<u>Q-values</u>	S1	S2	a1	-5	0	a2	-5	0	4 M
<u>Q-values</u>	S1	S2									
a1	-5	0									
a2	-5	0									
Q. B5	Answer following questions in 2-3 sentences. i. When we should use multiple-linear regression? ii. How regularization prevents overfitting? iii. How to find values of parameters which will help to reduce cost across iteration? iv. How Hidden Markov model differs than Markov model?	4 M									

***** ALL THE BEST *****

Question Paper

COURSE : Machine Learning (CS F464)

COMPONENT : Test 2 OPEN BOOK

DATE : 22/04/2021

WEIGHTAGE : 20% (20 Marks)

DURATION : 50 Minutes (4.30pm to 5.20pm)

Note: Please write down any assumptions that you make.

Q. 1 Consider a following plotted data points in Figure 1,

4 M

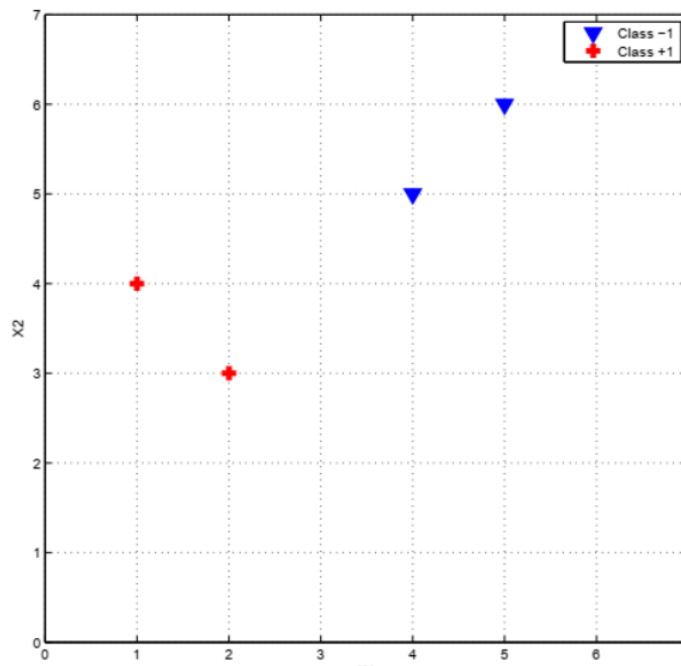


Figure 1 Q1

Find the support vectors and draw the decision boundary which will discriminate the given points into two classes. (Show step by step process of calculating a hyperplane)

Q. 2. Given a following datapoints and decision stumps. Find out the strong classifier (after two iterations) which will classify the data points into two classes.

6 M

Class A data points (X, Y)	<i>G (1,4), H (1,2), D (6,5), E (6,1)</i>
Class B data points (X, Y)	<i>C (3,3), F (3,1)</i>
Decision stumps	<i>$X < 2$, $X < 4$, $X > 7$</i>

Q. 3. Suppose Mr. X purchased share A on first day, on next day there is 60% chance of purchasing share A, a 30 % chance of purchasing share B, and a 10% chance of purchasing share C. On a day when Mr. X purchased share B, the next day's he will purchase share A is 40%, a 30% chance of purchasing share B, and a 30% chance of purchasing share C. Lastly, on any specific day when Mr. X purchased share C, there is a 20% chance of purchasing share A, a 50% chance of purchasing share B, and a 30% chance of purchasing share C on a following day. So, by

3 M

considering Mr. X purchased share A today, calculate the probability that Mr. X will purchase share A two days from now.

- Q. 4.** Suppose we have the following training set of positive (+) and negative (-) instances and a single test instance (o). All instances are projected onto a vector space of two real-valued features (X and Y) as shown in Figure 2. Answer the following questions. **3 M**

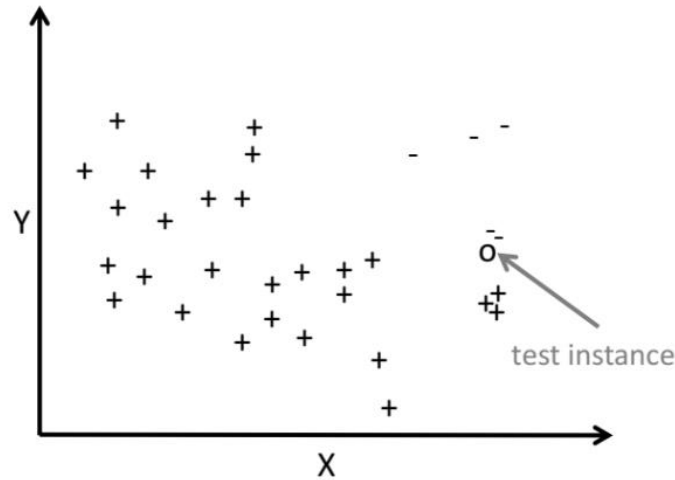
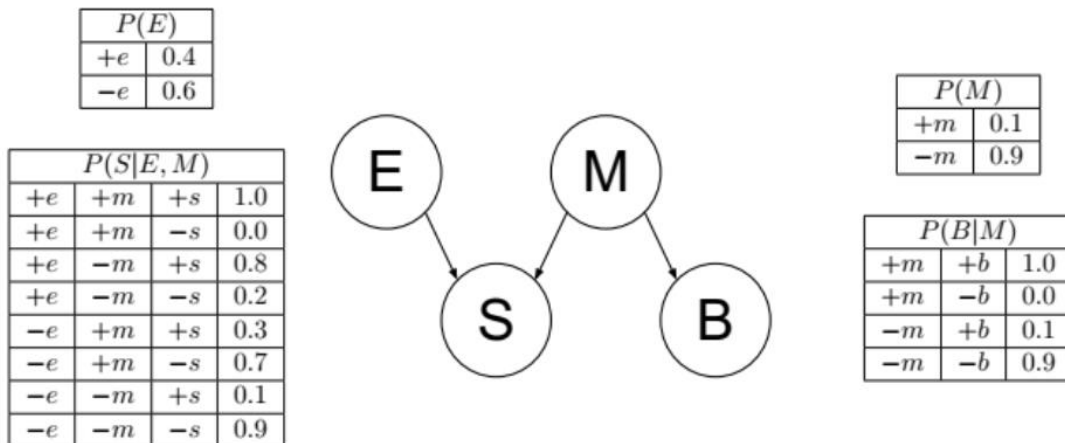


Figure 2 Q4

- What would be the class assigned to this test instance for $K=3$?
- Can setting $K=11$ is good for a given dataset? Give correct justification for your answer.

- Q. 5.** Consider the following Bayesian network and corresponding conditional probability tables (as shown in Figure 3). Calculate the below probabilities considering the given Bayesian network, **4 M**



***** ALL THE BEST *****

Question Paper

COURSE : Machine Learning (CS F464)

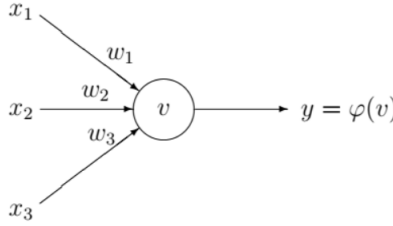
COMPONENT : Test 1 CLOSED BOOK

WEIGHTAGE : 20% (20 Marks)

DATE : 11-March-2021

DURATION : 50 Minutes (7:00 pm-7:50 pm)

Note: Please write down any assumptions that you make.

Q. 1	Consider a following confusion matrices <table><tr><td>55</td><td>5</td></tr><tr><td>10</td><td>30</td></tr></table> Calculate precision, recall and F1 score .	55	5	10	30	2 M																								
55	5																													
10	30																													
Q. 2	Consider the following set of training examples, <table><tr><th>Instance</th><th>Classification</th><th>A1</th><th>A2</th></tr><tr><td>1</td><td>+</td><td>T</td><td>T</td></tr><tr><td>2</td><td>+</td><td>T</td><td>T</td></tr><tr><td>3</td><td>-</td><td>T</td><td>F</td></tr><tr><td>4</td><td>+</td><td>F</td><td>F</td></tr><tr><td>5</td><td>-</td><td>F</td><td>T</td></tr><tr><td>6</td><td>-</td><td>F</td><td>T</td></tr></table> What is the information gain of A2 relative to these training examples?	Instance	Classification	A1	A2	1	+	T	T	2	+	T	T	3	-	T	F	4	+	F	F	5	-	F	T	6	-	F	T	2 M
Instance	Classification	A1	A2																											
1	+	T	T																											
2	+	T	T																											
3	-	T	F																											
4	+	F	F																											
5	-	F	T																											
6	-	F	T																											
Q. 3	Indicate following statements are True or False. Correct the statement if false. 1. Discriminant functions are linear combinations of the predictor or independent variables, which will best discriminate between the categories of the criterion or dependent variable. 2. To project the data to a larger dimension and to avoid class overlapping, fisher linear discriminant maintains a smaller variance among the dataset classes and a small variance within each of the dataset classes.	2 M																												
Q. 4	Which kind of learning is called as lazy learning? Why it is called as lazy learning?	2 M																												
Q. 5	Consider the below single artificial neuron,  Suppose that the weights corresponding to the three inputs are w1=2, w2=-4, w3=1 and the activation function $\varphi(v) = \begin{cases} 1 & \text{if } v \geq 0 \\ 0 & \text{otherwise} \end{cases}$. Calculate the output value y of the neuron for following set of inputs.	2 M																												

	<table><tr><td>Input</td><td>P1</td><td>P2</td></tr><tr><td>x1</td><td>1</td><td>0</td></tr><tr><td>x2</td><td>0</td><td>1</td></tr><tr><td>x3</td><td>0</td><td>1</td></tr></table>	Input	P1	P2	x1	1	0	x2	0	1	x3	0	1																																																																
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x1	1	0																																																																											
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x3	0	1																																																																											
Q. 6	Consider a following data for car age (x) and price (y), <table><tr><td>Car Age (in years)</td><td>Price (in dollars)</td></tr><tr><td>4</td><td>6300</td></tr><tr><td>4</td><td>5800</td></tr><tr><td>5</td><td>5700</td></tr><tr><td>5</td><td>4500</td></tr><tr><td>7</td><td>4500</td></tr><tr><td>7</td><td>4200</td></tr><tr><td>8</td><td>4100</td></tr><tr><td>9</td><td>3100</td></tr><tr><td>10</td><td>2100</td></tr><tr><td>11</td><td>2500</td></tr><tr><td>12</td><td>2200</td></tr></table> (a) Calculate the values of Θ_0 and Θ_1, and decided the equation of the linear regression line ($y = \Theta_0 + \Theta_1x$). Show step by step calculation. (4 M) (b) Use the regression line equation to predict the price of car if age of car is 9 years. (1 M)	Car Age (in years)	Price (in dollars)	4	6300	4	5800	5	5700	5	4500	7	4500	7	4200	8	4100	9	3100	10	2100	11	2500	12	2200	5 M																																																			
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12	2200																																																																												
Q. 7	Consider a below dataset of 14 training examples of the target concept “Buys_ computer”, where each instance is described as attributes Age, Income, Student, Credit_rating. <table><tr><td>Age</td><td>Income</td><td>Student</td><td>Credit_rating</td><td>Buys_computer</td></tr><tr><td><=30</td><td>high</td><td>no</td><td>fair</td><td>no</td></tr><tr><td><=30</td><td>high</td><td>no</td><td>excellent</td><td>no</td></tr><tr><td>31 to 40</td><td>high</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>>40</td><td>medium</td><td>no</td><td>fair</td><td>yes</td></tr><tr><td>>40</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>>40</td><td>low</td><td>yes</td><td>excellent</td><td>no</td></tr><tr><td>31 to 40</td><td>low</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td><=30</td><td>medium</td><td>no</td><td>fair</td><td>no</td></tr><tr><td><=30</td><td>low</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>>40</td><td>medium</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td><=30</td><td>medium</td><td>yes</td><td>excellent</td><td>yes</td></tr><tr><td>31 to 40</td><td>medium</td><td>no</td><td>excellent</td><td>yes</td></tr><tr><td>31 to 40</td><td>high</td><td>yes</td><td>fair</td><td>yes</td></tr><tr><td>>40</td><td>medium</td><td>no</td><td>excellent</td><td>no</td></tr></table> Apply Navie Bayes Classifier on given training data to classify the novel instance X = (age <=30, Income = medium, Student = yes, Credit_rating = fair) in a correct class. Show the steps by step procedure of calculating all posterior and prior probabilities to classify the instance in a correct class (Training and Testing Phase).	Age	Income	Student	Credit_rating	Buys_computer	<=30	high	no	fair	no	<=30	high	no	excellent	no	31 to 40	high	no	fair	yes	>40	medium	no	fair	yes	>40	low	yes	fair	yes	>40	low	yes	excellent	no	31 to 40	low	yes	excellent	yes	<=30	medium	no	fair	no	<=30	low	yes	fair	yes	>40	medium	yes	fair	yes	<=30	medium	yes	excellent	yes	31 to 40	medium	no	excellent	yes	31 to 40	high	yes	fair	yes	>40	medium	no	excellent	no	5 M
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